ESHA HOODA

SF Bay Area • esha.hooda@sjsu.edu • 669-288-3776 • LinkedIn • GitHub

Seeking Internships in ML/AI/Data Science and Product Management

EDUCATION

Bachelor of Science, Data Science, Minor in Statistics, Concentration in Public Health - San Jose State University Dec 2027

Coursework: **Machine Learning**, **Artificial Intelligence**, Statistics & Probability, Data Visualization, Data Structures & Algorithms, Database Management, Advanced Python, OOP, Linear Algebra, Discrete Math, UNIX, Technical Writing | Planned: Computer Vision, R, Social Network Analysis, Stats for Bioinformatics, Contemporary Environmental Health Issues, Health Issues in a Multicultural Society

Certificate: **Fundamental AI Algorithms and Applications in Computer Science** (expected completion: May 2027)

SKILLS & CERTIFICATIONS

Product & Data Tools: Figma, **Sketch**, **Jira**, **Linear**, **Google Analytics**, **Optimizely**, **SQL**, **Python**, Tableau, Power BI, Excel, Data Analysis, A/B Testing, Hypothesis Testing

Technical Skills: Python, **PySpark**, **ETL**, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch, **GCP**, **Git**, Flask, MySQL, SQLite

AI Tools & Applications: Anthropic Claude API (tool-calling, agent design), **Claude Code**, **GitHub Copilot**, ChatGPT, Gemini, Perplexity, Grok AI (used for research, task automation, and prompt engineering)

Certifications: Python, Machine Learning, Artificial Intelligence (IIT Bombay & IIT Madras), Code in Place (Stanford), IRB Research Training, CRLA Level 1 & 2

Languages: English (Native), Hindi (Native)

TECHNICAL EXPERIENCE

Deep Learning Intern - National University of Singapore & Hewlett Packard Enterprise Dec 2022 - Jan 2023

- Designed and implemented ML and deep learning models including ANN, RNN, GAN, SVM, NLP, and time series architectures; gained hands-on experience with Azure deployment pipelines
- Trained and optimized a posture detection model achieving 93% accuracy through neural network architecture tuning and iterative experimentation

Learning Assistant - Peer Connections, SJSU Jan 2025 - Jan 2026

- Mentored 250+ students in math and CS through **strong communication** and cross-functional collaboration with professors and peer tutors to co-design equity-focused in-class learning activities; earned CRLA Level 1 & 2 certification

PROJECT EXPERIENCE

Stock Volatility Prediction Pipeline May 2026

Tools Used: Python, **PySpark**, **SQL**, **ETL**, Pandas, Matplotlib, Seaborn, **GCP**, UNIX, Git

- Engineered a **distributed data pipeline** using **PySpark** on GCP processing 600,000+ daily trades across 500+ stocks (2019-2023); RandomForest feature importance identified prior-day intraday range as the dominant volatility predictor
- Served as both **lead engineer and PM** across several sprints on a student team: **defined the feature roadmap**, **drove project vision**, and **aligned stakeholders** through weekly report-outs and dashboards built for different audiences, taking full ownership of both technical execution and product direction throughout
- Confirmed COVID-era (Mar-Dec 2020) volatility was 50-80% above baseline; output written in partitioned Parquet for efficient querying

AI Job Application Tracker Agent Aug 2026

Tools Used: Python, Flask, SQLite, **Anthropic Claude API (tool-calling)**, JavaScript, HTML/CSS. Planned: RAG (embeddings + vector search)

- Identified a real gap in my own job search (manually re-reading postings and comparing them to my resume) and **scoped, built, and shipped** a full-stack agent to solve it end-to-end, rather than a fixed script: given a job posting, it **autonomously decides which tools to call** (parse the posting, fetch the resume, score fit, log the result)
- Designed the scoring logic to **avoid inflating fit scores**, prioritizing an honest, useful product decision over a flattering one; five-tool architecture backed by a Flask API and SQLite database, effectively a **recommendation problem**: matching a candidate to the right opportunity
- Planned (this week): a **feature roadmap** for a **PM-facing view** of the project, tracking usage patterns and framing next-build decisions the way a stakeholder report would, not just shipping features one at a time

Gaming Stats Dashboard & Pricing Models Dec 2025

Tools Used: Python, NumPy, Pandas, Scikit-learn, Seaborn, Matplotlib, **Git**

- Developed a classification pipeline (Logistic Regression, Random Forest, SVM) on 20K+ games across Steam, PlayStation, and Xbox to inform **pricing decisions**; diagnosed a 2:1 **class imbalance masking true model performance**, revealing models caught only 16-18% of high-price games despite 69% overall accuracy
- Recommended F1/recall over accuracy for imbalanced targets based on diagnostic findings, the kind of metric-choice **judgment call** that changes what a team decides to act on

Fake News Detection Using LSTM & BERT May 2026

Tools Used: Python, **TensorFlow/Keras**, **PyTorch**, HuggingFace, scikit-learn, NLTK, Git

- Compared four NLP architectures (SVM, Random Forest, custom BiLSTM, fine-tuned BERT) on a 20,000-article dataset; identified and documented a **source-style confound** (real articles concentrated in Reuters) likely inflating all four models' scores, and recommended cross-source validation as follow-up
- Constructed a multi-step preprocessing pipeline (tokenization, stopword removal, sequence encoding); tuned batch size, learning rate scheduling, and dropout regularization to mitigate overfitting

Recipe Management Database System Dec 2025

Tools Used: MySQL, **Flask**, JavaScript, Flutter, **Git**

- **Scoped and shipped a full product end-to-end**, not just a model: a full-stack CRUD application with **normalized schemas**, **indexed queries**, and RESTful API integration, plus synthetic data generation (~30K rows) to test the product's analytics against realistic usage before real users ever touched it

RESEARCH EXPERIENCE

Student Researcher - Research Foundation, SJSU May 2025 - Present

- Applied Principal Components Analysis (PCA) and Factor Analysis (**scikit-learn**) to an 11-item Likert survey (n=619) to validate three subscales (Identity & Belonging, Academic Efficacy, Navigation & Resources)
- Conducting independent research quantifying the LA model's impact on student outcomes (focus: gender equity and underrepresented groups); manuscript currently underway, with findings communicated to faculty and stakeholders via **data visualizations and written reports**
- Building and maintaining research databases; cleaning messy institutional data from a 60,000-row student outcomes dataset for **statistical analysis** using **Python** and hierarchical linear modeling (HLM)

LEADERSHIP EXPERIENCE

Outreach Lead - Peer Connections, SJSU May 2025 - Jan 2026

- Scaled a major initiative from 0 to 3,000 participants in 2 months by **redesigning internal processes** and building SQL-based **tracking systems** while **coordinating a team of 120+**

Vice President - Math Club, SJSU Oct 2025 - Mar 2026

- Organized weekly problem-solving events and partnered with faculty to promote engagement in mathematics

AWARDS & ACHIEVEMENTS

- State Rank 2 (3 consecutive years) in IEO, with international ranks of 232, 474, and 660 among millions of students across 70+ countries
- Top 150 nationwide, National Children's Science Congress (Govt. of India; selected from 1 million students)